

Can Large Language Models Count in Legal Czech? Subject–Predicate Distance as a Probe of LLM Reliability for Rule-Based Legislative Drafting Support

Anonymous ACL submission

Abstract

Tools that support legislative drafting increasingly need to check *structural* rules—limits on paragraph counts, on cross-reference nesting, or on other measurable properties of a sentence. We ask whether large language models (LLMs) can serve as the engine for such rules, using a narrow, objectively scoreable proxy: identifying the main subject (*podmět*) and predicate (*přísudek*) of a Czech legal sentence and counting the linear word-distance between them. We evaluate two family-matched open model pairs (Llama-3.1-8B / Llama-3.3-70B and Qwen-2.5-7B / Qwen-2.5-72B) under three prompting regimes—raw text, self-generated CoNLL-U, and gold CoNLL-U supplied—on a stratified sample of 640 sentences from the KUK 1.0 corpus, each run three times at temperature 0 (23,040 generations). Structure helps and larger models are better, but no configuration is reliable: the best condition—the 72B model *handed* the gold parse—is still only 65% exact. A decomposition shows the residual failure is *counting*, not structure-reading: given the parse the large models name the correct subject and predicate 83–87% of the time, yet even when they name both words correctly they still miscount the distance 28–36% of the time; the small models miscount almost always. The instability is severe (up to 56% of triplets vary run-to-run even at temperature 0 with a pinned provider) and majority voting does not fix it. A parser-quality control against a human gold treebank shows the silver oracle agrees with humans 99.3% on the labels, so these failures are real. We conclude that structural drafting rules should be implemented as deterministic extractors over a dependency parse ($\approx 99\%$ accurate and zero-variance) rather than as LLM prompts.

1 Introduction

Software that assists in drafting or reviewing legislation is beginning to encode explicit *rules* for text quality. Some of these rules are semantic (“is this

provision unnecessarily convoluted?”); others are *structural* and involve exact measurement—e.g. a limit on the number of paragraphs per section, on the nesting depth of cross-references, or on other complexity metrics derived from sentence structure. A natural implementation question is whether a single LLM prompt can serve as the engine for both kinds of rules, or whether structural rules require a different mechanism.

Independently, a growing body of work shows that LLMs are unexpectedly unreliable at *exact counting*. Performance swings by large margins with tokenization (Zhang et al., 2024), models fail at seemingly trivial character counts (Fu et al., 2024), and counting has been proposed as a minimal probe of rule-following reliability (Dai and Fan, 2026). If LLMs cannot reliably count in a controlled setting, they are a poor foundation for counting-based drafting rules.

We make this concrete with a deliberately narrow task in Czech legal text: given a sentence, identify its main-clause subject and predicate and compute the number of words that lie between them. The task is a proxy—not a drafting rule itself—but it is objectively scoreable, it is directly tied to an established syntactic-complexity measure (dependency distance; Gao and He, 2024), and it isolates counting from heavy semantics. We use it to answer three questions: (RQ1) can LLMs identify the subject/predicate pair? (RQ2) can they count the distance? (RQ3) does supplying explicit syntactic structure (a CoNLL-U parse) help, and does the effect depend on model size?

Our contributions are: (i) a fully reproducible, provider-pinned evaluation of four open models across three prompting regimes on 640 stratified legal sentences; (ii) a decomposition separating *role identification* from *counting*, which reframes the “can LLMs count” question; (iii) a quantification of run-to-run instability and a demonstration that majority voting does not remedy it; (iv) a parser-

quality control against a human gold treebank establishing that the silver oracle is trustworthy on the labels; and (v) a concrete recommendation for structural drafting rules. Code, prompts, the sampled evaluation set, and all model traces are released.

2 Related Work

LLM counting and rule-following. Zhang et al. (2024) show tokenization alone causes large accuracy swings on counting; Fu et al. (2024) document persistent failures on character counting. Dai and Fan (2026) frame stable counting as a minimal probe of rule-following reliability. Our task instantiates this probe on natural legal text with a linguistically motivated target.

Structured input for parsing-like tasks. Ginn and Palmer (2025) show that supplying symbolic/in-context rules improves LLM zero-shot dependency parsing, with the benefit shrinking as more examples are added. Our regimes 2 and 3 (self-generated vs. supplied CoNLL-U) test the analogous question for a structural measurement task.

Dependency distance as complexity. Linear distance between a syntactic head and its dependent is an established metric of syntactic complexity and working-memory load (Gao and He, 2024; Liu, 2008). Subject–predicate distance is arguably the single most load-bearing instance, which grounds our metric as more than an ad-hoc token count and ties it to the comprehensibility concerns of legislative drafting.

Czech legal NLP resources. We use the KUK 1.0 corpus of Czech legal/administrative text (Hladká et al., 2024), parsed with UDPipe 2 (Straka, 2018) into Universal Dependencies (Nivre et al., 2020), and the Czech Legal Text Treebank (Kříž and Hladká, 2018) (via its UD conversion) as a human-gold parser-quality reference.

3 Task and Data

3.1 Task formalization

Given a sentence, let p be the position of the main-clause subject and q the position of the predicate. The distance is

$$d = |p - q| - 1, \quad (1)$$

the number of tokens strictly between them in linear order (not tree depth). We operationalize the

subject as the single `nsubj/nsubj:pass` dependent of the sentence root, and the **predicate** as the head of that subject—i.e. the root—whatever its part of speech. This handles copular clauses correctly and matches the head–dependent pair that the dependency-distance literature measures: in “*Zákon je účinný*” (“The law is effective”) the predicate is the nominal/adjectival part *účinný*, with the copula *je* recorded separately.¹

Crucially, distance is counted in **whitespace-delimited words**, not CoNLL-U tokens. This is the human-meaningful unit and the only unit a model can count on raw text without being handed a tokenization standard; punctuation glued to a word and Czech split-tokens (e.g. *aby* → *aby+by*) each count as one word. We map UDPipe’s tokens to words using the CoNLL-U character offsets, so gold and model answers are commensurable in every regime.

3.2 Data

KUK 1.0 (Hladká et al., 2024) contains four sub-corpora of Czech legal/administrative text (ombudsman opinions, Frank Bold materials, ombudsman information flyers, and mixed advice texts), with silver UD parses from UDPipe 2 (czech-pdt-ud-2.15) and named-entity tags from NameTag 3 (Straková et al., 2019). It comprises 6,275 CoNLL-U files and 910,373 sentences. We treat UDPipe’s parse as the silver gold; §5.3 quantifies how far this can be trusted.

Filtering. We keep sentences with exactly one root, exactly one root-attached `nsubj` (no coordinated/clausal subjects, no multiple finite clauses), and word length in $[5, 80]$. This retains **415,462 / 910,373 (45.6%)** sentences, consistently across sub-corpora—itself a descriptive statistic about how often “simple” subject–predicate structures occur in legal Czech.

Evaluation sample. From the filtered pool we draw a stratified sample of **N=640**, crossing four sentence-length buckets with four distance buckets (4×4) and allocating an *equal* 40 sentences per stratum. This deliberately over-samples the long/far tail (population median distance is 1; sample median is 4.5), where difficulty concentrates. The draw is balanced across the four sub-corpora (160

¹This choice is provisional pending confirmation by a linguist; the copula index is logged so the alternative is recoverable without re-running.

each) so results are not implicitly a claim about one genre, and is fully seeded for reproducibility.

4 Experimental Setup

Regimes. Each sentence is presented under three regimes: (1) **raw text**—the model reads the sentence and returns subject, predicate, and distance; (2) **self-CoNLL-U**—the model first emits its own CoNLL-U parse, then the answer; (3) **CoNLL-U given**—the model is handed UDPipe’s parse and only needs to locate *nsbj*, follow its head, and count. All prompts are in Czech with two few-shot examples (one verbal, one copular) identical across regimes.

Models. We use two family-matched pairs so size is manipulated within an architecture: Llama-3.1-8B and Llama-3.3-70B (Dubey et al., 2024), and Qwen-2.5-7B and Qwen-2.5-72B (Qwen Team, 2025). All are served at fp8 through a single pinned provider per model, so quantization is held constant and the run is reproducible.

Protocol and metrics. Each (sentence $\times$ regime $\times$ model) cell is run three times at temperature 0, one sentence per call—23,040 generations. We report: format validity (a parseable answer); *role identification* (subject/predicate word-index match); *distance* exact-match, ± 1 , ± 2 , and MAE; and *consistency* (fraction of 3-run triplets that are identical). Full request/response traces, including the provider-reported per-call cost, are stored; the total run cost was USD 7.73.

5 Results

Table 1 presents the main results. We organize the discussion around four hypotheses.

H1: structure helps. Exact-match rises from raw to CoNLL-U-given for three of four models, and strongly for the large ones: Llama-70B 20.9 $\rightarrow$ 59.6 and Qwen-72B 25.3 $\rightarrow$ 64.9. Using paired McNemar tests on the large models, every pairwise regime difference is significant, and raw vs. CoNLL-U-given is overwhelming ($p < 10^{-45}$). The value lies almost entirely in *being handed* the parse, not in generating it (self-CoNLL-U helps only modestly, $p \approx 10^{-8}$ to 0.009).

H2: larger is better. Within both families and every regime, the larger model wins by a wide margin (e.g. CoNLL-U-given: 59.6 vs. 4.2 for Llama; 64.9 vs. 23.9 for Qwen).

Model	Regime	Exact	± 1	Pos.
Llama-8B	raw	8.0	29.5	5.2
	self-CoNLL-U	8.2	40.7	18.0
	CoNLL-U given	4.2	51.6	35.4
Qwen-7B	raw	7.4	20.3	2.8
	self-CoNLL-U	12.2	42.8	21.3
	CoNLL-U given	23.9	51.5	32.1
Llama-70B	raw	20.9	42.7	10.6
	self-CoNLL-U	26.1	43.8	20.1
	CoNLL-U given	59.6	67.5	50.8
Qwen-72B	raw	25.3	49.4	15.5
	self-CoNLL-U	39.1	56.3	32.0
	CoNLL-U given	64.9	71.1	51.6

Table 1: Distance exact-match, ± 1 -tolerance, and **Pos.** (both subject and predicate reported at their correct word position) (%), by model and regime, over 640 sentences $\times$ 3 runs. §5.1 separates *naming* the right words from *positioning* them. Format validity was $\geq 93\%$ throughout ($\geq 99\%$ except Llama-8B self-CoNLL-U).

H3: does structure *compensate* for size? No—if anything it *widens* the gap. The raw-text size gap is ≈ 13 –18 points; with CoNLL-U given it is ≈ 41 –55 points. Structure lets large models convert the parse into a correct count; the small models cannot.

H4: degradation with distance. Exact-match falls monotonically with the gold distance. Pooled over models, the CoNLL-U-given rate across distance buckets {0-1, 2-4, 5-9, 10+} is 48 $\rightarrow$ 51 $\rightarrow$ 39 $\rightarrow$ 15; raw text is 26 $\rightarrow$ 22 $\rightarrow$ 11 $\rightarrow$ 3. The long tail—exactly the adversarial case for legal Czech—collapses in every regime.

5.1 Finding the words vs. counting the span

The best condition—Qwen-72B handed the gold parse—is still only 64.9% exact. It is tempting to read this as the model failing to *read* the supplied structure, but the residual failure is mostly *counting*. We separate two abilities: *naming* the correct subject and predicate words, and *counting* the distance between them (Table 2).

Given the parse, the large models *name* the right subject and predicate most of the time (both correct 83–87%), yet their distance is exact only 60–65%. Even restricting to generations where they named *both* words correctly, the distance is exact only 63.8% (Llama-70B) and 72.3% (Qwen-72B): handed the parse *and* the right two words, the large models still miscount the span ≈ 28 –36% of the time. Llama-8B essentially never produces the correct distance even when it named both words

Model (CoNLL-U given)	Names both	Dist. exact	Exact both named
Qwen-72B	82.7	64.7	72.3
Llama-70B	86.7	59.6	63.8
Qwen-7B	41.0	23.5	38.4
Llama-8B	45.5	4.2	0.1

Table 2: With the parse supplied: how often the model *names* both the correct subject and predicate words, its overall distance exact-match, and the exact-match *restricted* to the cases where it named both words correctly (all %). Even given the parse *and* the right two words, the large models still miscount $\approx 28\text{--}36\%$ of the time; Llama-8B essentially never counts correctly.

(0.1%).

The error is specifically in *counting word positions*. The large models report the correct predicate word 93–97% of the time but its correct linear position only $\approx 55\%$ of the time (subject: named 87–88%, positioned 73–75%); the endpoint deeper in the sentence—usually the predicate—is mis-positioned far more often, the signature of a count that drifts with length. In a typical failure the model names the right words (e.g. *provozovatel, hradi*) but places them at positions 30 and 17 instead of 25 and 14, giving distance 12 instead of 10. In short, with the structure supplied, identifying *which* words are subject and predicate is largely solved; what remains unreliable—even for the 72B model—is the counting itself.

5.2 Instability

Even at temperature 0 with a pinned provider and quantization, the three runs disagree on **36%** (raw text) and **56%** (self-CoNLL-U) of sentence $\times$ model triplets; CoNLL-U-given is far more stable (11%). Two points matter for deployment. First, the instability *tracks difficulty*: for raw text the fraction of varying triplets grows with distance (29% $\rightarrow$ 41%). Second, and decisively, **majority-vote-of-three does not improve accuracy**—it equals the single-run rate in every cell (e.g. Qwen-72B raw 25.3 $\rightarrow$ 24.8; Llama-70B CoNLL-U-given 59.6 $\rightarrow$ 60.0). When the model varies, it flip-flops between *wrong* answers, so self-consistency or resampling will not deliver repeatable rule outcomes. A *consistent* answer can also be consistently *wrong*: Llama-8B given CoNLL-U is 80% consistent but 4% correct—reproducibility is not correctness.

5.3 Is the silver oracle trustworthy?

Because we score against UDPipe, we re-parse 740 human-gold sentences from the Czech Legal Text Treebank (Kříž and Hladká, 2018) with the *same* UDPipe model and compare. Where UDPipe commits to a comparable single-clause parse (295/740), it agrees with the human gold on subject (100%), predicate (99.3%), and *distance-exact* (99.3%), MAE 0.14. The silver *labels* are thus highly reliable on legal Czech, and the model failures above are real, not oracle noise. The silver noise that does exist is in *sentence segmentation*: for the remaining 445 sentences UDPipe split the gold sentence into two or more, disagreeing about where a sentence ends rather than about its subject and predicate.

6 Discussion: implications for drafting tools

Two results together are decisive for how structural rules should be built. First, the LLM’s weak link is the counting itself: even handed the gold parse *and* having named the correct subject and predicate, the best model still miscounts the distance $\approx 28\%$ of the time (§5.1)—whereas computing the distance from two identified endpoints is deterministic arithmetic, exact in code. Second, our noise-floor shows a dependency parser identifies those endpoints $\approx 99\%$ correctly on legal Czech (§5.3). Hence a **deterministic extractor over the UD parse would be $\approx 99\%$ accurate and zero-variance**, against $\leq 65\%$ and highly unstable for the best LLM prompt. Standard LLM mitigations—temperature 0, provider pinning, majority voting—demonstrably do *not* yield repeatable counts (§5.2).

We therefore recommend that legislative-drafting tools **triage each rule**: structural / countable rules (paragraph counts, reference nesting, dependency-distance-style complexity) should be implemented as deterministic functions of a parse, *not* as LLM prompts; “LLM + structured context” is warranted only for rules requiring semantic judgment a parser cannot express. The one caveat that carries over to any method is sentence segmentation, which on legal Czech is parser-dependent and must be treated as a fallible, first-class step upstream of any counting rule.

7 Conclusion

On a narrow, objectively scoreable proxy—subject-predicate word-distance in Czech legal text—no

LLM configuration we tested is reliable, the best (a 72B model handed the gold parse) reaching only 65% exact. Larger models read the structure reasonably well but still miscount the span $\approx 30\%$ of the time even when handed the parse and having named the correct endpoints; smaller models cannot count at all; and the run-to-run instability cannot be tuned away. Because a dependency parser identifies the endpoints $\approx 99\%$ correctly and the count is then deterministic, structural drafting rules are best served by deterministic extraction over a parse rather than by prompting an LLM.

Limitations

The evaluation is silver-standard: we measure agreement with UDPipe, not linguistic truth. We mitigate this with a human-gold parser-quality control (§5.3) showing 99.3% label agreement, but the segmentation disagreement it surfaces is a genuine limitation for end-to-end deployment. The operationalization of “predicate” as head-of-subject is a defensible but provisional choice awaiting linguist confirmation. We test one prompt formulation per regime and do not perform prompt-engineering sweeps; results may shift with different phrasings. The four models are all open-weight and $\leq 72\text{B}$; whether the counting bottleneck shrinks at frontier scale is an open question we did not test. Finally, success or failure on this proxy predicts, but does not prove, (un)reliability on other counting-type rules (paragraph counts, reference depth); we frame it as a probe, not a general claim about “LLMs and counting.”

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